BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

The Internet, DNS and Cloud Storage

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does a NIC provide?

2. Valid ages are 13-18. Which is normal test data?
   A) 12
   B) 19
   C) 'twelve'
   D) 15

3. For a numeric age input, give erroneous data of the wrong type.

4. A login asks for a username, password and another factor. "Authentication proves all the user's input is correct." Correct this. What else needs checking?

Today's objectives

Define the Internet as a worldwide network of networks. Describe how DNS finds an IP address for a domain name.

Key vocabulary - Read, cover, recall

Internet
Worldwide system of connected networks.

DNS
Domain Name System; resolves names to IP addresses.

Domain name
Readable name used to identify an Internet service.

IP address
Address used for routing data on networks.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

The Internet
The Internet is a worldwide set of interconnected computer networks where clients can request resources from servers.

DNS
The Domain Name System uses distributed servers to look up IP addresses associated with domain names.

Cloud storage and software
Cloud storage puts files on remote servers; cloud software and processing also provide services using remote computing resources.

Benefits and limitations
Cloud services support access from different devices, but depend on connectivity, service availability and secure access.

I Do - Worked model

A pupil enters a website's domain name in a browser. Describe how DNS helps them reach its web server. [4 marks]

- The pupil begins with a readable domain name, not the server's numeric network address.
- The device queries a DNS service to find the matching IP address [1].
- DNS looks up the domain [1] and returns the IP address to the client [1].
- The browser connects to the web server using that address [1]. Check the sequence.

We Do - Guided practice

1. A visitor enters the name of a local museum website. Describe the DNS lookup and connection in four stages. [4 marks]

2. A student stores revision notes on a remote service. Explain one advantage and one disadvantage. [4 marks]

3. Explain the difference between a client and a server when watching a web-hosted tutorial. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write one line to output the demonstration IP address.

```
domain = "museum.example"
ip = "192.0.2.7"
```

You Do - Independent exam questions

Q2. State one purpose of DNS. [2 marks]

Q3. A charity keeps its shared donation records in cloud storage but loses its Internet connection during an event. Explain a benefit and a limitation of cloud storage for this charity. [4 marks]

Q4. A visitor enters the domain name of a community sports club's online ticket page. Describe how DNS, the visitor's client device and the web server work together to load the page. [6 marks]

Exit Ticket - True or False?

DNS associates human-readable domain names with IP addresses.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The interface that enables a device to connect to a network.

2. D - 15

3. The text 'twelve' is the wrong data type and must be handled or rejected.

4. Authentication checks evidence of a user's identity. It does not prove their input is correct or safe.

I Do - Explain the model

The pupil begins with a readable domain name, not the server's numeric network address.

The device queries a DNS service to find the matching IP address [1].

DNS looks up the domain [1] and returns the IP address to the client [1].

The browser connects to the web server using that address [1]. Check the sequence.

We Do - Model answers

1. Client asks DNS for an address [1]; DNS finds an IP address linked to the name [1]; returns the IP address [1]; browser connects to the web server [1].

2. Notes can be accessed from different connected devices [1], enabling work at home or school [1]. Internet/provider availability is needed [1], so outages can block access [1].

3. The student's device is a client [1] requesting the tutorial [1]; the remote server hosts/provides the content [1] and responds to the request [1].

Code First - Answer

```
domain = "museum.example"
ip = "192.0.2.7"
print(ip)
```

Expected output: 192.0.2.7

You Do - Self-mark

Q2: It looks up an IP address [1] associated with a requested domain name [1].

Q3: Remote donation records are accessible from different connected devices [1], helping staff use records at events and at headquarters [1]. Cloud access requires an available Internet connection/provider [1], so an outage may prevent the charity from retrieving those records [1].

Q4: Client starts with name [1] and requests DNS lookup [1]; DNS locates address [1] and returns IP [1]; client contacts server [1], which returns requested content [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.